INCH-POUND

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February 28, 1997
SUPERSEDING
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30 September 1986

COMMERCIAL ITEM DESCRIPTION

TAPES, MEASURING: TANK GAUGING TYPE, WITH REEL AND PLUMB BOB

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers reel-mounted, steel measuring tapes, with plumb bob, for gauging the level and temperature of liquids in tanks.

1.2 Intended use. The measuring tapes are used for gauging the level of liquids in tanks, including but not limited to fresh water, sea water and oils including fuel and lubricating oil.

2. CLASSIFICATION

2.1 Classification. The tapes shall be of the following types, sizes and tape finish as specified (see 7.1):

Type I - Plumb bob for measuring liquids
Type II - Plumb bob for measuring liquids and temperature

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5210

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Size 1 - 25 feet
Size 2 - 50 feet
Size 3 - 75 feet

Tape finish I - Light background
Tape finish II - Satin - chrome white

3. SALIENT CHARACTERISTICS

3.1 Design and construction. The reel-mounted measuring tape, with plumb bob, shall consist of the components shown on figure 1. When uncoiled, the tape shall be removable from the reel and the reel from the frame without the need of tools. The tape assembly shall be capable of withstanding the load test specified in 5.3.1.

3.1.1 Frame assembly. The frame assembly shall consist of the frame, nonconductive handle lockable winding arm, and an adjustable tape wiper with a replaceable pad. It shall be corrosion resistant, durable, serviceable and permanently secured.

3.1.2 Tape assembly. The tape assembly shall consist of the tape reel, spindle and measuring tape; it shall be durable; corrosion-resistant; free-moving and easily serviced for tape replacement.

3.1.3 Tape. The tape shall be commercial spring-tempered steel; nominal 1/2 inch wide; and 0.008 to 0.013 inches thick. The tape shall be fitted with a swivel snap to accommodate a plumb bob; have a reinforced end to minimize bending, and be capable of coiling without causing permanent set when tested as specified in 5.3.2.

3.1.3.1 Tape marking. The tape shall have black markings on a light background (tape finish I) or a satin chrome-white finish (tape finish II); one sided graduation lines indicating feet, inches and eighth of inches; and raised or recessed numerals and graduations.

3.1.4 Tape refills. When specified (see 7.1) refill tapes of specific sizes meeting the requirements of 3.1.3 and 3.1.3.1 shall be furnished.

3.1.5 Tape accuracy. The overall tape length, exclusive of the first foot, shall be measured at an ambient temperature of $68^{0}F \pm 2^{0}F$. When supported on a horizontal surface and subjected to a tension load of 10 pounds, the measurement shall not exceed an error of 0.125 inch in 100 feet when tested as specified in 5.3.3.

3.1.6 Plumb bob - type I. The plumb bob shall be brass; 1 inch at the largest diameter; approximately 6.75 inches long; tapered at one end with a flattened point not to exceed 0.156 inches

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diameter; and weight rated at 20 ounces. The plumb bob shall have a serrated area approximately 3/16 inch wide by the bob length and an attachment point for the swivel snap measuring 6.600 inches from the flattened point to the attachment point (upper edge of hole).

3.1.7 Plumb bob - type II. The plumb bob shall be brass; shall have a cross section encompassing a 1 inch diameter; 6.75 inches long; tapered at one end and weight rated at 15 ounces. The plumb bob shall also have a 6.600-inch scale graduated in 1/4 inch increments; an attachment point for the swivel snap measuring 6.6 inches from the point to the attachment point (upper edge of hole); and hollowed out on one side to form an oil cup and to house a thermometer with a range from minus 20^0F to plus 120^0F.

3.1.8 Plumb bob accuracy. The contact point in the swivel snap (for plumb bob, inside end of lower snap) shall be 6.600 inches $\pm$ 0.016 inch. The error, including plumb bob and swivel snap from zero to the 1-foot graduation, shall be not greater than 0.031 inch when tested as specified in 5.3.4.

3.1.9 Plumb bob replacement. When specified (see 7.1), plumb bobs meeting the requirements of 3.1.6 or 3.1.7 and 3.1.8 shall be furnished.

3.1.10 Grounding strap. The measuring tapes can be obtained with an optional grounding strap (see 7.1 and figure 1.)

3.2 Marking. Identification, warning and instruction marking shall be legible and permanently inscribed or attached on the frame of the end item. Numbering and lettering shall be no less than 1/16 inch in height.

3.2.1 Identification. The end item shall be marked with the manufacturer's name, type and tape length.

3.2.2 Warning. The end item shall have a warning notice containing but not limited to the potential of shock/explosion hazards due to static electricity.

3.2.3 Instructions. The end item, as required for the optional grounding strap, shall contain instructions on the proper sequence of operation while utilizing the ground strap.

3.3 Workmanship. The end item shall be clean and smooth on all surfaces. The handles shall have no splits, seams, cracks, splinters, or gouges. Painted or enameled surfaces shall have no uncoated area, no chipped film and no discoloration. All metal parts shall be free of sharp edges, fins, burrs, splits, and machine marks. Plating, when applicable, shall show no oxide scale, rust, roughness, or foreign inclusions and shall be smooth and uniform with no uncoated, nonadherent, or peeling areas.

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3.4 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this commercial item description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS

4.1 Contractor-recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Market acceptability (MA). The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 2 years.

b. The company must have sold 1000 units meeting the requirements of this CID, in the commercial or government marketplace over the past 2 years.

5.3 Tests.

5.3.1 Load test. The assembled tape and reel shall be tested in a locked position, with no less than 1 foot of tape extended and with a 20-pound load hanging from the swivel snap. The 20-pound load shall then be removed and the tape allowed to uncoil full length by the weight of the plumb bob. Any evidence of breakage, permanent distortion, binding, or failure of the locking mechanism shall constitute failure of the test.

5.3.2 Coiling test. The tape shall be rolled around a rod having a nominal diameter of 1 3/4 inches and then removed. Any evidence of permanent set shall constitute a failure.

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5.3.3 Tape accuracy test. The measurement shall be conducted at an ambient temperature of $68^0F \pm 2^0F$. The uncoiled tape shall be supported on a horizontal surface, and a 10-pound load applied to the swivel snap. The accuracy shall be checked against a master rule, instruments, or laboratory equipment of known accuracy. Any error exceeding that indicated in 3.1.5 shall constitute failure of the test.

5.3.4 Plumb bob accuracy test. The measurement shall be conducted at an ambient temperature of $68^0F \pm 2^0F$. The tape with plumb bob attached shall be sufficiently unwound and supported on a horizontal surface. The accuracy of the zero to 1-foot graduation (including plumb bob) shall be checked against a master rule, instruments, or laboratory equipment of known accuracy. Any error exceeding that indicated in 3.1.8 shall constitute failure of the test.

6 PACKAGING

6.1 Packaging. Preservation, packing and marking requirements shall be as specified in the contract or purchase order (see 7.1).

7 NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents (if applicable):

a. Title, number and date of this CID
b. Type, size and tape finish required (see 2.1)
c. When only refill tapes including swivel snaps are required (see 3.1.4)
d. When only plumb bob replacements are required (see 3.1.9)
e. When optional grounding strap is required (see 3.1.10)
f. Preservation, packing, and marking required (see 6.1)

7.2 Sources for documents.

7.2.1 Sources for non-government association documents.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 380 - Use of the International System of Units (SI) (The Modernized Metric System)

(Application for copies of ASTM standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

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7.3 Sources of supply. Manufacturers whose products are known to meet the requirements of this CID, in whole or in part, are listed below; however, competition is not limited to these companies.

U.S. Tape Company
217 River Avenue
Patchogue, NY 11772

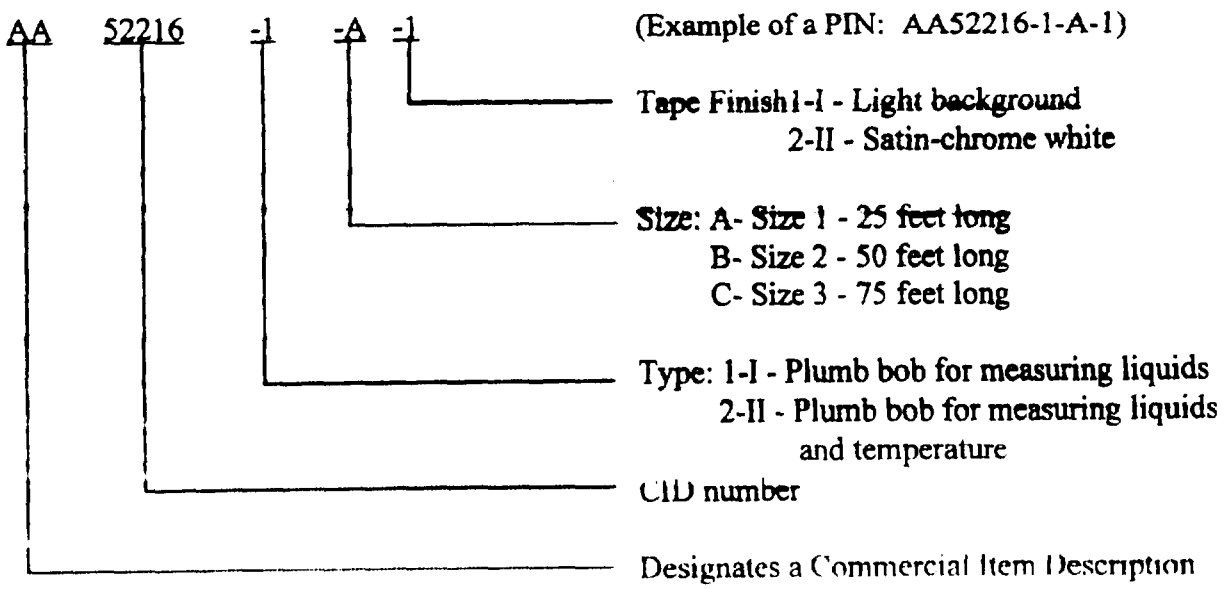
Cooper Industries
1000 Lufkin Road
P. O. Box 728
Apex, NC 27502

L.S. Starrett
121 Crescent Street
Athol, MA 01331

7.4 National Stock Number (NSN). The following is a list of NSNs assigned which correspond to this document. The list may not be indicative of all possible NSNs associated with the document.

NSN	Type	Size	Finish	Item
5210-00-221-1885	I	1	I	Tape, measuring
5210-00-221-1886	I	2	I	Tape, measuring
5210-00-221-1887	I	3	I	Tape, measuring
5210-00-526-0752	II	N/A	II	Tape, measuring, tank level gauge
5210-00-238-1734	I	N/A	N/A	Plumb bob
5210-00-293-1739	I	1	I	Tape, measuring, replacement
5210-00-293-1740	I	2	I	Tape, measuring, replacement
5210-00-347-2591	I	3	I	Tape, measuring, replacement

7.5 Part identification number (PIN). The PINs to be used for items required by this CID are created as follows:



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MILITARY INTERESTS:

Custodians

Army - GL
Air Force - 99

Review Activities

Army - AT
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

ARMY - GL

Project 5210-0294

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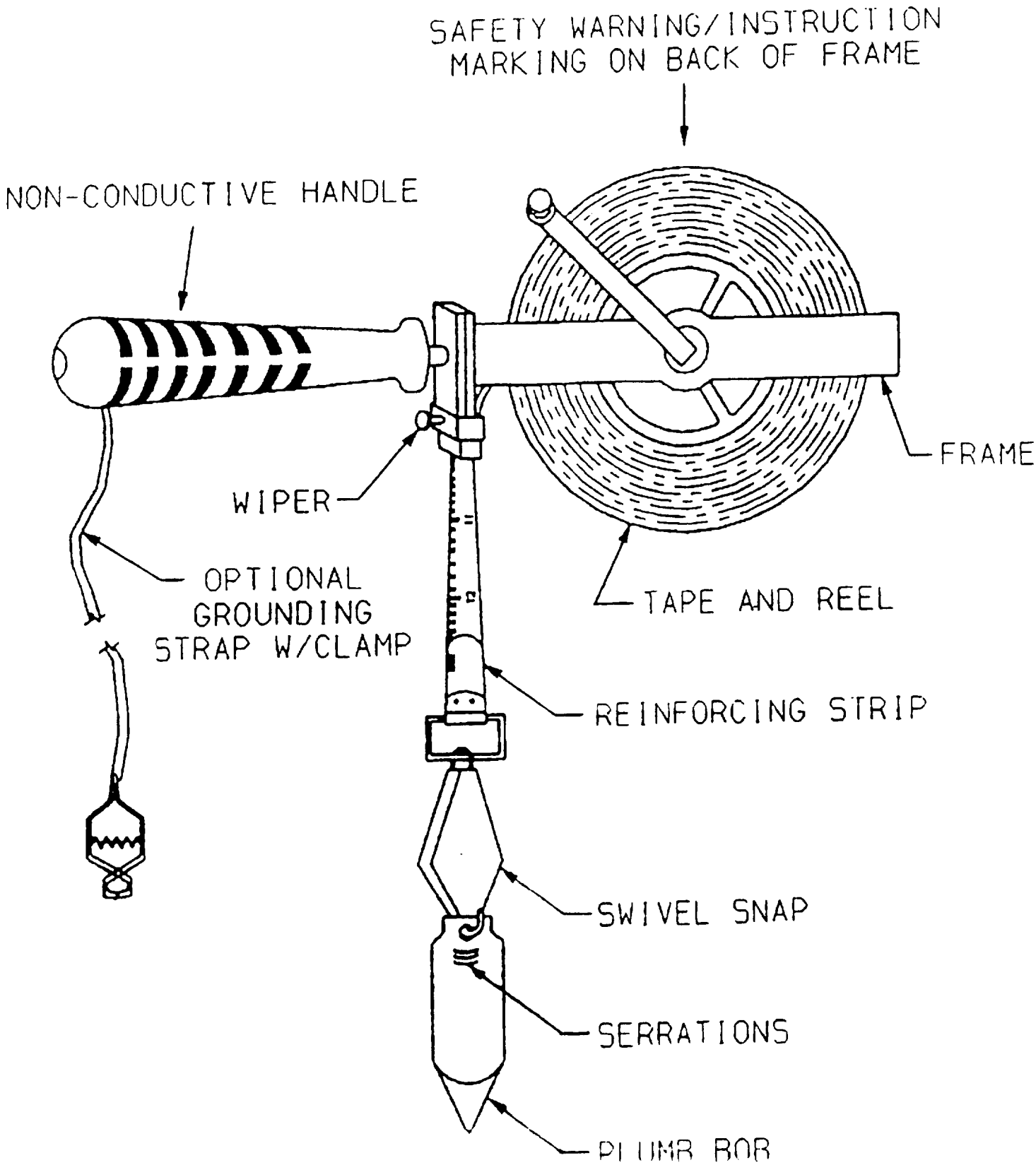


FIGURE 1. Tank gauging type measuring tape with reel and plumb bob

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMENDED CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-52216	97-02-28

3. DOCUMENT TITLE Tapes, Measuring: Tank Gauging Type, with Reel and Plumb Bob

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

[redacted]

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial ~~508-233-5175~~	(2) AUTOVON ~~256-5175~~
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center Attn: SSCNC-WEF Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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